

EarthDial: Turning Multi-sensory Earth Observations to Interactive Dialogues

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Abstract

Automated analysis of vast Earth observation data via interactive Vision-Language Models (VLMs) can unlock new opportunities for environmental monitoring, disaster response, and resource management. Existing generic VLMs do not perform well on Remote Sensing data, while the recent Geo-spatial VLMs remain restricted to a fixed resolution and few sensor modalities. In this paper, we introduce EarthDial, a conversational assistant specifically designed for Earth Observation (EO) data, transforming complex, multi-sensory Earth observations into interactive, natural language dialogues. EarthDial supports multi-spectral, multi-temporal, and multi-resolution imagery, enabling a wide range of remote sensing tasks, including classification, detection, captioning, question answering, visual reasoning, and visual grounding. To achieve this, we introduce an extensive instruction tuning dataset comprising over 11.11M instruction pairs covering RGB, Synthetic Aperture Radar (SAR), and multispectral modalities such as Near-Infrared (NIR) and infrared. Furthermore, EarthDial handles bi-temporal and multi-temporal sequence analysis for applications like change detection. Our extensive experimental results on 44 downstream datasets demonstrate that EarthDial outperforms existing generic and domain-specific models, achieving better generalization across various EO tasks. Our source codes and pre-trained models are at <https://github.com/hiyamdebary/EarthDial>.

1. Introduction

Recent advancements in VLMs enable unified visual interpretation, where a single model can perform diverse tasks such as classification, localization, visual question-answering, counting, visual reasoning, and visual ground-

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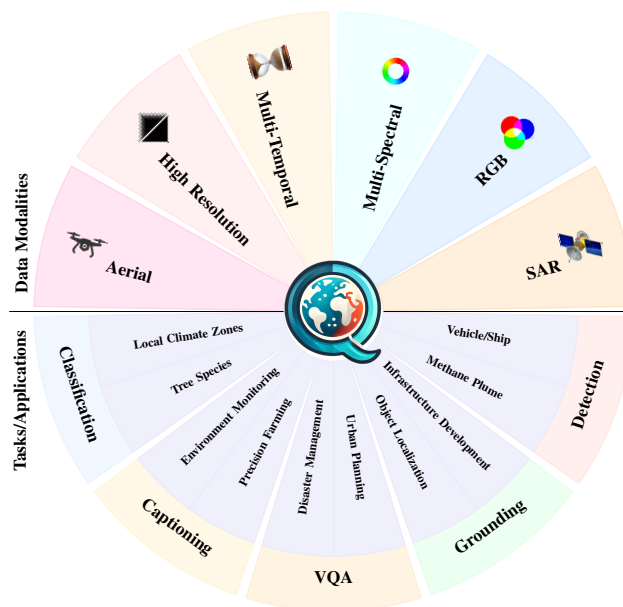


Figure 1. EarthDial is the first domain-specific VLM for earth observation data that can comprehensively interpret multi-sensor imagery. Specifically, our model covers visible RGB, SAR, multi-temporal, high-res satellite and aerial imagery available in varying spatial resolutions (*top half*). We develop the largest remote sensing image-text instruction dataset with over 11M samples. EarthDial can perform several multimodal understanding tasks: classification, detection, captioning, visual question-answering (VQA), and grounding (*bottom half*). This unlocks a number of downstream applications where EarthDial shows promising results.

ing [2, 10, 11, 33, 44, 46]. However, these generic VLMs do not scale well to Earth Observation (EO) data, which require specialized capabilities to handle the complex geospatial, spectral, and temporal dimensions of remote sensing (RS) data. Even state-of-the-art proprietary models like GPT-4V show low accuracies in domain-specific RS data [58], emphasizing the need for EO-specialized VLMs.

Recently, domain-specific VLMs have been developed

Dataset	Type	# Samples	OS	MS	MT	MR	Tasks															
							IC	RC	VQA	SC	MLSC	TSC	OD	VG	DA	BTCD	MTCD	M-TC	UHI	LCZ	TSC	MPD
RSICap [22]	Optical	2.6K	✓	×	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
VHM [38]	Optical	180K	×	×	×	×	✓	✓	✓	✓	✓	×	×	✓	×	×	×	×	×	×	×	×
VRSBench [28]	Optical	205K	✓	×	×	×	✓	×	✓	×	×	×	×	✓	×	×	×	×	×	×	×	×
GeoChat [24]	Optical	380K	✓	×	×	×	✓	✓	✓	✓	✓	×	×	✓	×	×	×	✓	×	×	×	×
MMRS [59]	Optical, SAR, IR	1.01M	✓	×	×	×	✓	✓	✓	✓	×	×	✓	✓	×	×	×	✓	×	×	×	×
SkyEye-968k [57]		0.97M	✓	×	×	×	✓	×	×	×	×	×	×	✓	×	×	×	✓	×	×	×	×
LHRS-Instruct [36]	Optical	81K	✓	×	×	×	✓	×	×	✓	✓	×	×	×	×	×	×	✓	×	×	×	×
FIT-RS [35]	Optical	1.8M	✓	×	×	×	✓	✓	✓	✓	×	×	✓	✓	×	×	×	✓	×	×	×	×
EarthDial-Instruct (ours)	Optical, SAR, S2, IR, NAIP	11.11M	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Table 1. Overview of RS VLM datasets, their types, and supported tasks. OS: Open-Source, MS: Multi-Spectral, MT: Multi-Temporal, MR: Multi-Resolution. Tasks: IC: Image Captioning, RC: Region Captioning, VQA: Visual Question Answering, SC: Scene Classification, MLSC: Multi-Label Scene Classification, TSC: Temporal Scene Classification, OD: Object Detection, VG: Visual Grounding, DA: Disaster Assessment, BTCD: Bi-Temporal Change Detection, MTCD: Multi-Temporal Change Detection, M-TC: Multi-Task Conversation, MPD: Methane Plume Detection, UHI: Urban Heat Island, TSC: Tree Species Classification, LCZ: Local Climate Zones. Our dataset is 6× larger and covers diverse sensing modalities and provides annotations for a rich set of downstream tasks.

to understand EO data using generative multimodal models. RS-GPT fine-tuned the MiniGPT-4 model on 2.5k remote sensing instructions [22]. GeoChat represents an initial effort capable of performing image and region-level understanding as well as visual grounding in high-resolution remote sensing images [24]. Several approaches have focused on data scaling; e.g., LHRS-Bot uses crowd-sourced labels from OpenStreetMap to obtain 1.15M RS image-text pairs for multimodal alignment [36]. SkyEyeGPT curates a 968K sample instruction-following dataset for remote sensing conversational tasks [57]. However, these efforts are limited in high-resolution image processing and do not support multi-spectral, multi-temporal analysis.

In this work, we present EarthDial, aiming to develop the first unified model that can cohesively process multi-resolution, multi-spectral, and multi-temporal remote sensing imagery to unlock numerous downstream tasks. In all the above modalities, EarthDial can perform diverse tasks, including classification, object/change detection, question-answering, image and region captioning, and visual grounding. To achieve this goal, we propose the most extensive instruction tuning dataset to date, with over 11.11M instructions covering visible imagery with varying resolutions, SAR, and multispectral modalities, including NIR and infrared. Furthermore, EarthDial can handle bi-temporal and multi-temporal sequence analysis for applications such as change detection and temporal sequence classification. Our main contributions are as follows:

- We propose EarthDial, a conversational VLM capable of processing multi-spectral, multi-temporal, and multi-resolution remote sensing imagery with natural text queries, addressing a wide range of EO tasks.
- We introduce the largest instruction tuning dataset for remote sensing, comprising over 11.11M instruction pairs across various modalities, enhancing the model’s understanding and generalization capabilities.
- Experimental results demonstrate that EarthDial performs well in comparison to existing domain-specific VLMs, achieving higher accuracies and better generalization

across 44 downstream EO tasks.

2. Related Work

Generic Vision-Language Models (VLMs): The development of generic VLMs like VisualGPT [9], BLIP [27], Flamingo [3], and Kosmos [23] has enabled advancements in showcasing multi-modal understanding by aligning visual and language data for diverse applications. Devoted efforts from researchers enabled the VLMs to perform a range of tasks, for example, OCR to diagram and infographics understanding to video analysis within a unified model [11, 26, 29, 46]. The continuous progress enabled the alignment of additional modalities such as audio, video, 3D point clouds [18, 45], audio-video grounding tasks [15], 3D visual grounding [51] as well as in fields such as LIDAR [30, 61] and robotics [20, 49]. Nevertheless, they struggle with the unique contextual complexities of remote sensing (RS) data, which requires specialized alignment for geospatial, spectral, and temporal information.

Geospatial VLMs: Recently, various efforts have been devoted towards domain-specific RS vision-language understanding to address the limitations of general VLMs [22, 24, 32, 36, 57, 59]. RemoteCLIP [32] employs contrastive learning over RS image-text pairs, illustrating the zero-shot classification and image-text retrieval capabilities. RS-GPT [22] fine-tuned over EVA-CLIP and Vicuna LLM demonstrates image captioning and VQA abilities while struggling over detection and visual grounding tasks. GeoChat [24], LHRS-Bot [36] and SkyEyeGPT [57] extend their capabilities to resolve multiple tasks such as region-level understanding as well as visual grounding in high-resolution RS images. However, these models do not cover multi-spectral and temporal modalities. More recently, EarthGPT [59] introduces the MMRS1M dataset to integrate optical, SAR, and infrared modalities, advancing multi-sensor RS comprehension. However, they do not cover other multi-spectral inputs and lack generalization to multi-temporal and varying resolution inputs.

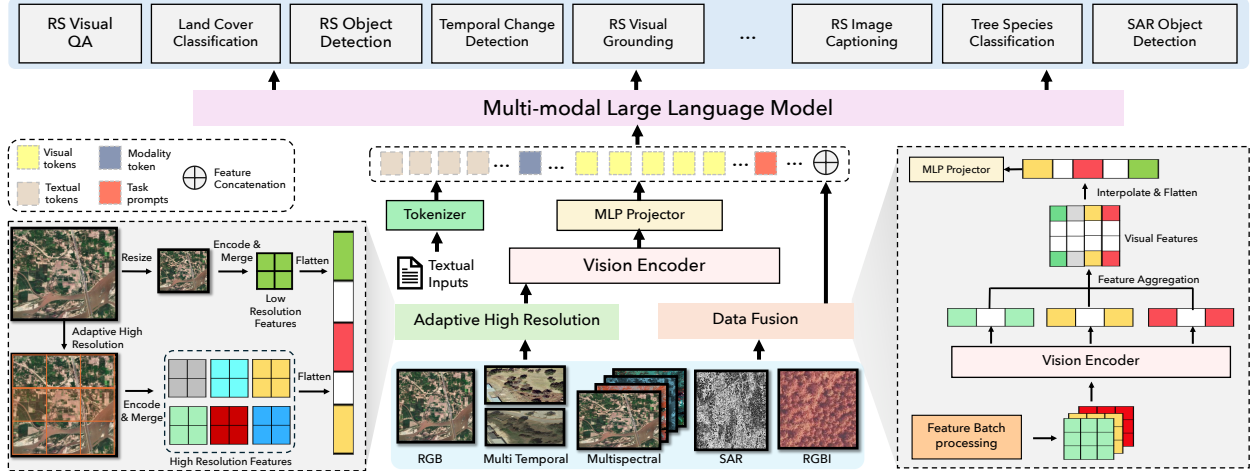


Figure 2. EarthDial Architecture: The model can take a diverse set of inputs ranging from RGB to multi-spectral and time-series images. Multi-resolution inputs are converted to tokens based on an adaptive high-resolution block [11] that includes both local and global features. The multi-channel inputs (multi-spectral/temporal) are converted to tokens via the data fusion block, which aggregates features across all channels. The resulting visual tokens are mapped to LLM input space using MLP projectors and concatenated with the textual inputs. We use special task and modality tokens to distinguish between several input modalities and downstream tasks (Table 2). The LLM is trained with multimodal inputs to perform a number of downstream tasks, ranging from VQA to detection, grounding and change detection.

RS Instruction Datasets: A large number of instruction-following datasets have been introduced to train the RS-VLMs effectively. For example, GeoChat-Instruct [24], SkyEye-968k [57], and RS5M [60] provide extensive RS image-text pairs, supporting instruction-based VLM training on optical data. These existing datasets limit the model’s capabilities across different sensor modalities. On the other hand, MLLMs [59], provide image-text pairs from optical, SAR, and infrared images. Regardless of the large scale, the aforementioned datasets often lack diverse RS applications across various modalities, including multi-resolution, multi-spectral, and multi-temporal RS sensor data. The recently introduced dataset aims to enhance content richness with factual and deceptive questions, enabling VLMs to address a wider range of tasks in the RS domain [38]. However, there is no large-scale unified instruction-following dataset that can encapsulate the distinctive contextual complexities of diverse RS applications across different modalities, multi-resolution, multi-spectral, and multi-temporal RS sensor data. Our work is an effort to bridge this gap with an 11M instruction set to seamlessly integrate multiple earth observation modalities covering diverse spectral and with time-series imaging data for diverse RS applications.

3. EarthDial

Our goal is to develop a domain-specialized VLM that can handle complex geospatial, spectral, and temporal dimensions unique to RS imagery. As described above, the existing general and geospatial VLMs lack in understanding high-resolution, multi-spectral, and multi-temporal RS im-

agery. To bridge this gap, we propose a comprehensive large-scale instruction tuning dataset for RS domain with over 11M instruction, covering diverse resolutions and geographical locations. Building on this dataset, we propose EarthDial, the first unified model capable of processing multi-resolution, multi-spectral, and multi-temporal RS data across a variety of tasks, from classification and visual grounding to change detection.

EarthDial leverages state-of-the-art vision-language models (VLMs) for natural images and provides a multi-stage finetuning recipe to progressively expand model capabilities. Our model architecture builds on InternVL [11, 12] with specific modifications to enable multi-spectral and multi-temporal processing (Sec. 3.1). We proposed a three-stage model training process to enhance the model’s capabilities across multiresolution, multispectral, and multi-temporal datasets. We first pretrain with remote sensing datasets, focusing on adapting state-of-the-art VLMs for EO-specific dialogues. In the next stage, output of pretrained encoders and LLM are adapted using RGB and temporal imagery for downstream tasks. Finally, an extended finetuning stage is specifically designed to improve its performance with multispectral and synthetic aperture radar (SAR) datasets to broadly cover additional applications. Next, we explain our model design in detail.

3.1. Model architecture

As illustrated in Fig. 2, EarthDial consists of three trainable components: a visual encoder, an MLP layer projector, and a large language model (LLM). Our model is rel-

atively lightweight with only 4B parameters compared to the existing natural geospatial VLMs. The model is designed in such a way that it can take multi-resolution, multi-spectral, and time series datasets to generate various RS dialogues. As our visual encoder, we use InternViT-300M [12], a lightweight vision model distilled from the larger 6B InternViT that demonstrates strong visual encoding capability. Since our design goal is to have an efficient model, we use the Phi-3-mini pre-trained LLM [1]. To connect the visual encoder with the LLM, a simple MLP is used as the connector block to map visual tokens to the LLM space. As explained in Fig. 3, we tune the parameters of these three blocks systematically in different stages of training.

Furthermore, the model incorporates two key modules, Adaptive High Resolution and Data Fusion, that are crucial in applying EarthDial to different resolution inputs as well as multi-spectral and multi-temporal RS data.

Adaptive High Resolution: In remote sensing, images come in various sizes and resolutions, particularly high-resolution imagery where resizing for model can lead to the loss of critical pixel details. To address this, we adopt a dynamic resolution input strategy inspired by InternVL 1.5 [11], which enhances the model’s ability to capture fine-grained details. The approach dynamically selects an optimal aspect ratio from a set of pre-defined ratios, dividing the image into 448×448-pixel tiles and creating a thumbnail for global context to help the model understand the overall scene. Depending on the input resolution, 1 to 12 tiles can be created during training and upto 40 during inference. This approach minimizes aspect ratio distortion and accommodates varying resolutions during training and evaluation.

Data Fusion: The data fusion module in EarthDial is designed to improve the model’s capability to process multi-temporal, multi-spectral, synthetic aperture radar (SAR), and RGB/hyper-spectral datasets. For multi-spectral inputs of any type, it operates by iteratively processing three channels of data at a time, which are passed through the Vision Transformer (ViT), to extract features for each channel. The extracted features are then aggregated and reduced in size using bilinear interpolation via the AnyRes block [26] to ensure efficient handling of multi-spectral inputs. The AnyRes block splits the inputs into patches, encodes them, and uses bilinear interpolation to reduce tokens per patch, thus enabling the processing of multi-spectral inputs. These reduced visual embeddings are then concatenated with input corresponding text embeddings. The final step involves fusing these combined visual and textual features, which are passed to the LLM for further contextual processing. This fusion strategy allows EarthDial to integrate visual data from various modalities together with textual descriptions, improving its performance on complex RS tasks.

For RGB temporal images, we first pass each image through the ViT to extract visual tokens, then stack and con-

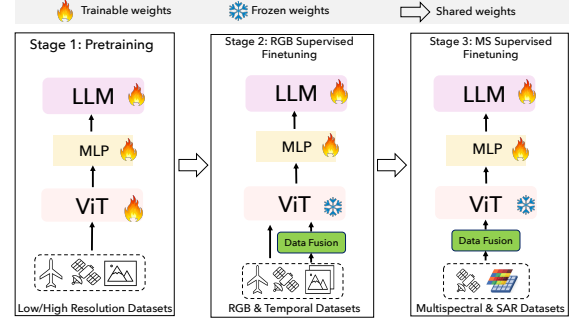


Figure 3. EarthDial training Strategy for different RS modalities. We first pretrain with RGB imagery of different resolutions to achieve better alignment. Thereafter only LLM and projectors are trained on RGB and temporal inputs. We then expand the model’s capability to multi-spectral and SAR imagery in Stage 3.

catenate these tokens before passing the combined representation to the LLM. Next, we explain our model training.

3.2. Stage-wise Model Training

EarthDial is initialized with pretrained backbones explained in Sec. 3.1. It is then trained in three distinct stages, designed to systematically expand model’s capabilities.

Stage 1 - RS Conversational Pretraining: The main goal of pretraining is to enhance the vision encoder’s ability to learn visual features from various satellite imagery sources, such as Sentinel-2, Landsat 8, Sentinel-1, and aerial images, and to generate descriptions for remotely sensed objects and scenes. During this phase, we use instruction datasets described earlier data section, specifically the Satlas and Skyscript datasets, which contain 7.6M image-text pairs for different types of remote sensing imagery.

In the pretraining stage, the objective is to establish alignment between multiple sensor modalities in the RS domain with their corresponding natural language text descriptions. This is typically achieved using large datasets of image-text pairs, such as image and captions. The model learns to predict the corresponding text for a given image in an autoregressive manner, optimizing performance through a standard cross-entropy loss function. This process helps the model improve its ability to generate accurate text descriptions from remote sensing visual inputs. As shown in Fig. 3, the architecture during pretraining is deliberately kept simple, only comprising of the vision encoder, MLP and the LLM, without any data fusion. At the pretraining stage, model is not introduced temporal and multi-spectral datasets, to keep the task relatively simple and initially learn strong representations from single-image RGB datasets from our proposed EarthDial-Instruct dataset.

At this stage, all the learnable components were trained to ensure proper alignment of the RS imagery. For pretraining the EarthDial model, we utilized 8 NVIDIA A100-80G GPUs. The training process employed an initial learning

rate of $4e-5$, optimized using a cosine learning rate scheduler. The key hyperparameters included the use of thumbnails to capture local features, an adaptive patch size ranging from 1 to 6 for capturing more detailed high-level features, a batch size of 2, and a maximum sequence length of 4096 tokens. Additionally, a weight decay of 0.01 was applied to regularize the model.

Stage 2: RS RGB and Temporal finetuning: In the second stage of model training, we carefully selected high-quality instruction sets with prompts aimed at teaching the model to better understand user commands and successfully complete desired tasks on different satellite imagery. This approach allows Large Language Models (LLMs) to generalize to unseen tasks, improving zero-shot performance across a variety of remote sensing tasks.

During this stage, we utilized previously pretrained model encoders and fine-tuned the MLP and LLM layers to perform visual instruction tuning for diverse remote sensing tasks, such as image captioning, classification, detection tasks (e.g., grounding, referring, identification), visual question answering (VQA), and temporal change detection.

For multi-temporal images, we applied data fusion techniques, as explained in the model overview section, before passing the data to the feature extractor. Additionally, we used all the RGB and time-series datasets described in Tab. 2 during this phase of model fine-tuning.

Stage 3: RS Multispectral and SAR Finetuning: In the third stage of model training, we extended the model’s capabilities to work with multispectral, high-resolution RGBI, and SAR images. This was achieved by introducing a data fusion module, as explained earlier, to handle imagery with more than three or fewer than three spectral bands. The goal was to enable the model to learn from multispectral and SAR data for various remote sensing tasks.

We utilized the pretrained weights from the previous stage 2 and fine-tuned the MLP and LLM layers, while keeping the ViT parameters frozen from Stage 1. The same ViT+MLP+LLM architecture was employed, with the addition of the data fusion module (Sec. 3.1) to integrate information from multiple spectral channels. This training stage allows the model to handle a broader range of tasks, including land cover classification, species detection, Methane plume detection, UHI and SAR-based ship detection. Training hyperparameter details are given in Table 3.

4. EarthDial-Instruct Dataset

Pretraining Instruction Data: With EarthDial, our primary aim is to improve generalization performance on diverse downstream tasks, covering a wide range of modalities, multi-resolution, and multi-temporal data. Therefore, we curate high-quality pre-train question-answer (QA) instruction pairs from SkyScript [48] and SatlasPretrain [6] data, which includes Sentinel-2 (S2), Sentinel-1 (SAR),

Stages	Datasets	Number of QA pairs	Token format
Stage 1	NAIP	3,000,113	[hr_rgb.0.5]
	Sentinel-2	2,749,511	[s2_rgb.10]
	Landsat	1,671,437	[l8_rgb.30]
	SkyScript	249,855	[s2_rgb.10]
Stage 2	Classification	565,853	[hr_rgb.0.5]
	Detection	22,624	[hr_rgb.0.5]
	Visual Grounding	17,845	[hr_rgb.0.5]
	Caption	202,530	[caption] [hr_rgb.0.5]
	VQA	630,768	[hr_rgb.0.5]
	Change Detection	64,631	[changedet][hr_rgb_temp.0.5]
	Disaster assessment	37,563	[hr_rgb_temp.0.5]
Stage 3	Geochat	308,861	[hr_rgb.0.5]
	Sentinel -1	1,668,043	[s1_vh.10]
	Local Climate Zones	765,591	[s2_ms.30]
	Tree Species	38,527	[treeclassify] [hr_rgb.0.5]
	Methane Plume	6,849	[hyper_rgb.3]
	Urban Heat Island	1,296	[uhi][l8_ms.30]

Table 2. Summary of the number of QA instruction pairs used during each stage, the image sources, and token formats.

Stages	QA pairs	Epoch	Train-time	GPUs	Batchsize	Grad. Accu.	Lr	Decay	Adaptive patch sizes
2	1.8M	1	4 (hours)	4A100 80G	2	64	$4e-5$	0.05	1 to 6
3	2.4M	1	6 (hours)	4A100 80G	2	64	$4e-5$	0.05	1 to 6

Table 3. Details of stages, hyperparams, & training duration

NAIP, and Landsat imagery along with labels. We choose InternLM-XComposer2 [19] for generating instructions using labels. Our data curation process involves filtering to ensure data quality. First, we filter out samples with sparse labels (<3). Second, we apply **luminance and coverage-based filtering** to remove cloudy and low spatial coverage images. Third, we prompt the InternLM-XComposer2 to generate QA instruction pairs based on the key attributes (points, polygons, object category, and position) specified in the inputs and labels. The details can be found in the supplementary material. The curated instruction stats across different imagery sources are presented in Tab. 2.

Downstream Tasks Image-text Instruction: While pre-training focuses on enhancing generalization capabilities, we also need task-specific fine-tuning with diverse data types to improve downstream performance. To handle this, we carefully curate a large number of instruction-following datasets that cover ten diverse downstream tasks (e.g., scene classification, object detection, visual question answering, image captioning, change detection, Methane plume detection, tree species classification, local climate zones, urban heat islands, and disaster assessment), six visual modalities (Optical, SAR, S2, Infrared, NIR, and Hyperspectral), and two visual temporal modalities (Optical and SAR). Details are in the supplementary.

Scene Classification: We construct scene classification instructions with nine standard scene classifications, one multilabel scene classification (BigEarthNet [41]), and one temporal scene classification (FMoW [16]). We limit the sequences to 4 images to handle multitemporal scene classification. We also use local climate zones (LCZ) [63] and TreeSatAI-Time-Series [4] datasets to determine the LCZs

and botanical tree species, respectively.

Object Detection: We curate instruction-following tasks utilizing three tags i.e., *refer*, *identify*, and *grounding* to perform region-level captioning, referring expressions, and grounded description for object detection datasets from various remote imaging modalities like optical, SAR, and infrared. We include visual grounding [42, 56] datasets for region-level captioning. Following [24], we compute the key attributes in the image such as the object’s category, bounding box, color, relative position, and relative size present in the detection dataset. We present the box as $[x_{min}, y_{min}, x_{max}, y_{max}, \theta]$. Here, (x_{min}, y_{min}) denotes the top left corner point while (x_{max}, y_{max}) presents the bottom right corner of the bounding box. The angle θ represents the rotation angle of the bounding box.

Visual Question Answering (VQA) & Image Captioning: We create VQA and image captioning instructions by including six VQA and five image captioning datasets.

Change Detection: We integrate three binary change detection datasets and one multitemporal (MUDS [52]) dataset. The original MUDS dataset has masks. Thus, to generate the instructions, we manually analyzed the images and masks and generated five captions for each sequence.

Methane Plume Detection: For the Methane plume detection, we utilize the STARCOP [40] dataset which presents labeled hyperspectral (groundtruth mask and emission rate) data. We conversationally prompt three questions; (i) is there any Methane plume present in the input, (ii) what is the location of the plume, and (iii) what is its emission rate?

Urban Heat Island (UHI): We compute land surface temperature (LST) and normalized difference vegetation index (NDVI) maps from S2 and Landsat imagery. Based on the data, we prompt to classify the underlying region into cooler, mildly hot, and extremely hot regions. We also generate instructions about what underlying land use caused the temperature and how to mitigate it.

Disaster Assessment: We use the xBD [21] dataset for building disaster assessment by utilizing bitemporal pre- and post-disaster images. We also include the QuakeSet [8] dataset and prompt to determine if an earthquake occurred between the bi-temporal SAR images and its magnitude.

5. Experiments

Here, we discuss the experimental results of the proposed EarthDial across a diverse set of applications, including RGB, multispectral, SAR, infrared, and thermal imagery. Our evaluation covers various tasks such as scene classification, referred object detection, region captioning, grounding descriptions, VQA, image captioning, change detection, and methane plume detection.

Scene classification: For zero-shot evaluation, we compare EarthDial with RGB datasets in Tab. 4, whereas BigEarthNet (RGB), xBD Set 1, and fMoW are supervised. We

Model	AID [50] (RGB)	UCMerced [53] (RGB)	WHU-19 [17] (RGB)	BigEarthNet [41] (RGB)	xBD Set 1 [21] (Temporal)	fMoW [16] (Temporal)
GPT-4o	74.73	88.76	91.14	49	67.95	21.43
InternVL-8B [12]	60.4	58.23	79.3	19.73	51.44	21.04
GeoChat [24]	72.03	84.43	80.09	20.35	53.32	59.2
EarthDial	88.76	92.42	96.21	68.82	96.37	70.03

Table 4. Comparison of classification accuracy across various datasets. EarthDial indicates a significant improvement in classification accuracy over other existing generic and specialized VLMs.

Method	BigEarthNet [41] (MS)	SoSAT-LCZ42 [63] (MS)	TreeSatAI [4] (RGBI)	Ship Dataset (SAR Imagery)				
				Small	Medium	Large	Single	Multiple
GPT-4o	49	15.53	16.73	0.70	0.90	3.20	1.20	0
EarthDial	69.94	60.72	56.61	12.14	26.02	35.56	26.03	6.06

Table 5. Performance evaluation of EarthDial across diverse modalities for multi-class classification and referred object detection tasks on SAR imagery. For MS modality, EarthDial achieves an average 32.5% improvement in classification accuracy compared to GPT-4o. For the RGBI modality, EarthDial achieves 40.2% higher accuracy than GPT-4o. A similar trend is observed for SAR imagery, where EarthDial delivers higher mAP@0.5, even when detecting multiple objects, highlighting the advantages of leveraging multi-modal inputs.

also compare multi-spectral (MS) datasets (BigEarthNet, SoSAT-LCZ42), and the RGB-Infrared TreeSatAI dataset as shown in Tab. 5. We notice that EarthDial shows consistent performance gain against existing generic and specialized VLMs. Moreover, EarthDial outperforms temporal scene classification FMoW as well as over xBD test-set 1 for disaster assessment [21] as in Tab. 4. Further results are in a supplementary document.

Object Detection: Following [24], we address three sub-tasks: referred object detection, region captioning, and grounding description. We consider existing generic VLMs like GPT-4o, InternVL2-4B while specialized GeoChat for the comparison. As existing InternVL2 doesn’t provide the rotated bounding boxes, for fair comparison, we finetune the InternVL2 on GeoChat-Instruct and compared it with our EarthDial. Table 6, 7, and 8 depict our EarthDial is a clear winner and consistently outperforms the all other compared VLMs by a large margin. Especially on grounding description tasks, where existing methods struggle to detect/localize the objects, our Earthdial achieves higher mAP. Also, improved results on SAR imagery datasets showcase the multi-modal data processing capability of our EarthDial. **Image Captioning and Visual Question Answering:** Our EarthDial outperforms the existing generic and specialized VLMs by clear margins over image captioning datasets as shown in Tab. 9. In addition, for the VQA task, we utilize datasets RSVQA-LRBEN and RSVQA-HRBEN (zero-shot) for evaluation, following [24]. Tab. 10 presents the VQA accuracy of existing models compared to the proposed EarthDial, outperforming most of the categories.

Change Detection: To demonstrate the temporal data processing capability of the proposed EarthDial, we evaluate its performance on a change detection task. We applied the data fusion strategy (discussed in Sec. 3.1) to merge tokens

Model	GeoChat-Instruct [24]					NWPU VHR-10 [13] (ZS)					Swimming Pool Dataset (ZS)					Urban Tree Crown Detection [55] (ZS)				
	Small	Medium	Large	Single	Multiple	Small	Medium	Large	Single	Multiple	Small	Medium	Large	Single	Multiple	Small	Medium	Large	Single	Multiple
GeoChat [24]	2.9	13.6	21.7	16	4.3	2.5	3.2	14.7	13.23	1.9	-	3.1	7.3	1.2	0.6	-	1.8	8.9	2.9	3.1
InternVL2-4B [12]	6.3	24.37	37.38	24.96	11.72	7.1	12.68	25.48	22.96	8.1	0.6	6.6	8.9	4.5	0.865	-	3.17	13.41	5.9	3.1
InternVL2-8B [12]	7.20	23.76	31.99	25.77	9.30	4.26	11.85	20.72	21.66	5.86	0.3	4.7	18.27	7.6	0.514	0.6	3.99	17.1	7.9	3.94
EarthDial	11.43	31.76	39.07	34.29	13.41	11.66	14.21	23.12	25.37	8.9	1.04	7.4	24.90	8.4	1.04	1.1	7.01	25.67	11.13	6.7

Table 6. Comparison of our EarthDial for referred object detection tasks across various datasets. Small, medium, and large denote the object size, while single and multiple denote the number of objects. Here, ZS means zero-shot evaluation.

Model	GeoChat-Instruct [24]			HIT UAV [43] (ZS)			NWPU VHR 10 [7] (ZS)			SAR-Ship Dataset [47]			SRSDD-v1.0 [25]			Swimming Pool (ZS)			UCAS AOD [62] (ZS)			Urban Tree Crown [55] (ZS)		
	R-1	R-L	MT	R-1	R-L	MT	R-1	R-L	MT	R-1	R-L	MT	R-1	R-L	MT	R-1	R-L	MT	R-1	R-L	MT	R-1	R-L	MT
GPT-4o	9.41	7.6	8.02	10.96	9.02	8.23	17.68	11.81	9.63	7.49	7.24	7.07	6.9	6.67	7.94	13.94	10.19	7.91	-	-	-	11.63	10.11	7.12
InternVL2-8B [12]	10.58	9.06	8.5	11	9.53	8.4	11.88	9.63	7.7	9.67	8.67	8.19	10.55	8.84	8.94	14.63	12	6.95	14.52	10.43	8.59	11.89	9.8	6.79
GeoChat [24]	72.77	72.74	61.9	59.85	59.85	51.31	65.02	65.02	53.31	57.15	57.15	52.2	63.72	63.72	57.31	64.73	64.73	51.47	65.03	65.03	52.4	60.52	60.52	50.48
EarthDial	73.38	73.34	62.72	61.83	61.83	52.80	72.14	72.14	60.01	63.1	63.1	54.83	68.8	68.8	62.45	61.96	61.96	47.42	64.03	64.03	52.82	63.47	63.47	54.09

Table 7. Comparison of our EarthDial with existing generic and specialized VLMs for region captioning task across various datasets.

Model	HIT UAV [43] (ZS)					NWPU VHR 10 [13] (ZS)					Swimming Pool Dataset (ZS)					UCAS AOD [62] (Zero-Shot)				
	@0.5	@0.25	R-1	R-L	MT	@0.5	@0.25	R-1	R-L	MT	@0.5	@0.25	R-1	R-L	MT	@0.5	@0.25	R-1	R-L	MT
GPT-4o	0.1	0.7	14.20	10.56	7.16	0.7	6.1	14.72	10.82	9.41	0.1	1.2	12.87	10.07	7.79	0.1	1.3	14.71	11.14	5.97
InternVL2-4B [12]	0.6	6.4	28.1	27.68	23.94	10.6	29.87	30.67	29.09	21.92	0.8	4.2	28.3	28.08	24.64	4.6	31.8	21.01	20.01	11.65
GeoChat [24]	0.8	8.0	22.82	22.22	22.27	2.2	15.27	21.46	20.74	21.38	1.8	8.8	21.45	21.15	23.94	1.45	13.63	20.02	18.81	14.22
EarthDial	2.61	13.86	28.31	28.06	22.25	17.07	41.00	27.05	26.35	23.12	1.9	7.4	29.7	29.31	22.77	8.5	34.02	21.17	20.28	13.01

Table 8. Comparison of EarthDial with existing generic and specialized VLMs on the grounding description task across multiple datasets.

Model	NWPU RESISC45 Captions [14]			RSCID Captions [34]			RSITMD Captions [54] (Zero-shot eval)			Sydney Captions [39]			UCM Captions [39]		
	Rouge1	Rouge-L	Meteor	Rouge1	Rouge-L	Meteor	Rouge1	Rouge-L	Meteor	Rouge1	Rouge-L	Meteor	Rouge1	Rouge-L	Meteor
GPT-4o	19.43	14.86	28.16	20.53	15.59	26.03	18.31	14.22	24.83	14.52	12.55	18.87	25.77	20.58	33.18
InternVL2-8B [12]	20.69	15.64	30.18	21.59	16.13	28.17	18.91	14.65	26.02	15.71	13.8	19.69	22.9	17.91	28.61
GeoChat [24]	14.86	12.54	15.21	13.48	11.59	12.39	13.41	11.5	12.33	12.0	11.26	10.63	14.4	13.22	14.27
EarthDial	45.84	39.96	80.61	33.77	27.61	56.18	26.74	21.72	34.06	49.39	41.0	57.31	40.0	34.15	51.42

Table 9. Comparison of our EarthDial with existing generic and specialized VLMs for Image captioning tasks across various datasets.

Model	Presence	Comp	R/U	Avg.	Model	Presence	Comp	Avg.
MiniGPTv2	55.16	55.22	39.00	54.96	MiniGPTv2	40.79	50.91	46.46
Qwen-VL [5]	38.57	67.59	61.00	55.35	Qwen-VL [5]	66.44	60.41	63.06
InternVL2-8B [12]	58.54	72.28	71.00	66.51	InternVL2-8B [12]	67.35	76.91	72.70
GeoChat [24]	91.09	90.33	94.00	90.70	GeoChat [24]	58.45	83.19	72.30
LHRs-Bot [36]	88.51	90.00	89.07	89.19	EarthGPT [59]	62.77	79.53	72.06
EarthDial	92.58	92.75	94	92.70	EarthDial	58.89	83.11	72.45

Table 10. Comparison of EarthDial with existing VLMs for visual question answering task (left: RSVQA-LRBEN, right: RSVQA-HRBEN). Comp: Comparison, R/U: Rural/Urban.

Model	Dubai CC			LEVIR MCI [31]			MUDS [52]			SYSU [37] (zero-shot)		
	R-1	R-L	MT	R-1	R-L	MT	R-1	R-L	MT	R-1	R-L	MT
GPT-4o	8.81	7.45	18.68	10.33	8.4	22.05	14.18	11.02	20.92	16.48	12.32	17.49
InternVL2-4B [12]	7.31	6.38	21.12	8.88	7.43	22.14	10.25	7.9	17.73	13.27	9.98	14.36
GeoChat [24]	14.21	14.19	28.91	17.15	35.42	12.35	12.28	12.23	15.98	13.45	12.02	13.96
EarthDial	31.94	30.66	55.83	33.78	30.47	74.8	28.16	24.03	33.56	18.03	17.42	14.98

Table 11. Comparison of EarthDial with existing generic and specialized VLMs on the change detection task.

obtained from multiple images across time. The results in Tab. 11 highlight EarthDial’s strong ability to interpret and respond effectively to temporal data.

Temporal Disaster Assessment: Here, we show the capability of our EarthDial in processing temporal data. First, we consider the benchmark xBD dataset belonging to the disaster assessment task for the experiment. xBD dataset has two images: pre-disaster and post-disaster. Thus, from xBD dataset, we prepare eight sub-tasks covering tempo-

ral image captioning, region classification, image classification, object detection, and referred object detection applications. Tab. 12 presents a summary of the performance results of both generic and specialized VLMs across the sub-tasks discussed above. EarthDial consistently outperforms all other existing VLMs by a significant margin, demonstrating its capability to effectively process temporal data for the desired task. Additionally, we evaluated our method on QuakeSet [8] for earthquake prediction using SAR imagery. We evaluate the model based on binary classification to determine whether an earthquake event occurred or not between the input SAR imagery. While GPT-4o achieves a classification accuracy of 55.86, our method outperforms it with an accuracy of 57.53.

Multi-modal Data Processing: To showcase the capability of EarthDial in processing multi-modal data, we examine multi-spectral (MS), RGB-infrared, and SAR imagery for classification and referred object detection tasks. Comparative results of the proposed EarthDial and existing GPT-4o are given in Tab. 5. Our EarthDial outperforms GPT-4o by a significant margin on both multi-spectral, RGBI, and SAR imagery. Significant improvement in the performance highlights the effectiveness of our multi-band fusion strategy.

Urban Heat Island: For UHI, we prompt to classify the

R-1, R-L, MT denotes ROUGE-1, ROUGE-L, METEOR scores

Model	Image Captioning			Region Classification		Image Classification			Object Detection		Referred Object Detection	
	R-1	R-L	MT	Test Set-1	Test Set-2	Test Set-1	Test Set-2	Test Set-3	mAP@0.5	mAP@0.25	mAP@0.5	mAP@0.25
GPT-4o	14.21	10.35	19.52	51.68	71.62	67.95	75.45	70.41	0.2	2.15	0	0
InternVL2-8B	13.89	10.37	14.92	14.39	58.33	51.44	61.52	51.12	0.6	1.07	0	0.7
GeoChat	14.18	10.67	12.20	25.30	57.65	53.32	52.19	49.51	1.15	7.2	0.2	3.09
EarthDial	87.26	87.26	88.53	53.7	83.09	96.37	82.85	54.01	7.6	21.11	5.1	13.09

Table 12. Comparison of our EarthDial for various tasks on the xBD dataset (temporal). R-1, R-L, and MT denote ROUGE-1, ROUGE-L, and METEOR scores, respectively. **Image Captioning:** Describes the damage observed in the post-disaster image. **Region Classification:** Test Set-1, model classifies the level of damage in the user-marked region. Test Set-2, binary classification of user marked region into 'damage' or 'no-damage' class. **Image Classification:** Test Set-1, classifies the image into the type of disaster. Test Set-2, binary classification of image into 'damage' or 'no-damage' class. Test Set-3, classifies the image based on affected building count (none, one, two, or many). Classification recall is used for the evaluation of Region and Image Classification tasks. **Object Detection:** Locates all large buildings in the post-disaster image. **Referred Object Detection:** Locates the user-referred damage region/object in the post-disaster image. It is clearly observed that our EarthDial significantly improves performance for all the tasks.

underlying region into cooler, mildly hot, and extremely hot regions. We achieve an accuracy of 56.77% in identifying the temperature trends from user-input Landsat8 bands, whereas GPT-4o achieves an accuracy of 22.68% in the same task. This shows the capability of our EarthDial in processing multi-modality data effectively compared to the existing generic GPT-4o.

Methane Plume Classification: We consider STARCOP dataset for the evaluation, which has a four-channel image (RGB + mag1c band). Our EarthDial processes these RGBM bands to predict the presence of methane plumes as "Yes" or "No". We compare our accuracy with the GPT4o model. We simply give RGB and Mag1c bands to the GPT4o and collect the response. With this, GPT4o achieves an accuracy of 40.93%, while our EarDial achieves 77.09% *i.e.*, improves it by 32.16%.

6. Ablation Study

Here, we demonstrate the effectiveness of our multi-stage pre-training, and multi-spectral band fusion strategy.

Effect of Multi-Stage Pre-Training: We assess the performance of the EarthDial model on a complex detection task, both before and after multi-stage pre-training. For this analysis, we focus on the referred object detection task using the GeoChat-Instruct dataset. With multi-stage pre-training, EarthDial's mAP @0.5 improves by 5%, showing a notable boost in detecting multiple referred objects compared to its performance without pre-training.

Effect of Multispectral Band Fusion: In this setup, we evaluate EarthDial's performance with respect to the data fusion module (using average pooling and bilinear interpolation) for multi-spectral band fusion, on the classification tasks for BigEarthNet and TreeSat AI. Table 13 shows that the bilinear interpolation fusion strategy is more effective than simple average pooling, enhancing model's average accuracy by 13.5%. The impact of multi-spectral data combined with our multi-band fusion strategy is evident in Table 5. EarthDial achieves a 1.75% improvement

Referred object detection task (GeoChat-Instruct dataset [24])								Multi-band classification		
AHR	Pre-training	Small	Medium	Large	Single	Multiple		Data Fusion	BigEarthNet MS	TreeSat AI
$\times$	$\times$	6.06	22.97	36.26	24.75	10.3		Average	47.66	49.09
$\times$	$\checkmark$	10.85	32.05	36.15	33.73	11.91		Max-pooling	34.62	29.84
$\checkmark$	$\checkmark$	11.75	33.33	42.60	34.78	15.03		Bilinear	67.01	56.93

Table 13. Ablation study about the effect of multi-stage training (sec. 3.2) and spectral band fusion strategy (sec. 3.1). The proposed multi-stage pre-training approach improves EarthDial's mAP @0.5 for detecting multiple referred objects by 5%. Additionally, the use of our bilinear fusion strategy enhances the average classification accuracy by 9.5%.

in classification accuracy on the Multi-Spectral (MS) version of BigEarthNet compared to its RGB counterpart. This demonstrates EarthDial's ability to leverage complementary information from multispectral bands, enhancing performance in multi-class classification tasks.

7. Conclusion

We present EarthDial, a conversational assistant purpose-built for Earth Observation (EO) data, capable of transforming complex, multi-sensory Earth observations into interactive, and natural language dialogues. EarthDial supports multi-spectral, multi-temporal, and multi-resolution imagery as input and addresses a wide spectrum of remote sensing tasks, including classification, detection, captioning, question answering, visual reasoning, and visual grounding. To enable this versatility, we developed a comprehensive instruction-tuning dataset containing over 11M instruction pairs, encompassing diverse modalities such as RGB, S2, Synthetic Aperture Radar (SAR), Near-Infrared (NIR), and infrared. Additionally, EarthDial excels in handling bi-temporal and multi-temporal sequence analysis, making it highly effective for applications like change detection/disaster assessment. Our experiments across 44 downstream tasks highlight EarthDial's superior performance over existing generic and domain-specific models, demonstrating its robust generalization capabilities and its potential to set a new standard in EO task automation.

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